

Scalability & Future Plan

Date	15 July 2026
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	1 Mark

This document outlines how Rising Waters can be scaled to handle larger user bases, increased data loads, or extended features, and captures the roadmap for evolving the project beyond its current state.

Current System Limitations

S · N o	Limitation	Impact	Priority
1	Cloud Cover is a derived proxy since the public rainfall dataset has no real cloud-cover column	Slightly reduces real-world prediction fidelity for that one feature	Medium
2	No live weather data ingestion; users must manually enter rainfall figures	Limits real-time usability compared to an auto-updating dashboard	High
3	Free-tier Render hosting causes a cold-start delay after inactivity	First request after an idle period can take 10–20 seconds	Low

Scalability Plan

S · N o	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single Flask instance on Render's free/standard tier; handles light concurrent traffic	Move to a paid Render tier with autoscaling, or run Gunicorn workers behind a load balancer
2	Data Storage	Flat joblib files for the model/scaler; CSV/Excel for training data	Migrate to a lightweight database (e.g. PostgreSQL) if user accounts, prediction history, or multi-region models are added
3	Performance	~0.4 sec average response time locally; XGBoost inference is lightweight	Add response caching for repeated identical inputs; pre-warm the hosted instance to avoid cold starts
4	Security	No authentication; public read-only prediction service, input validated server-side	Add rate limiting on /predict and optional API-key access if opened up for third-party integration

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1 (Current)	Core ML flood-risk prediction and public Flask web app	Completed — Jul 2026	Delivered a working, publicly accessible flood-risk tool
Phase 2	Live rainfall API integration (NASA POWER / Copernicus)	Target: Q4 2026	Removes manual data entry and enables real-time predictions
Phase 3	SMS / email early-warning alerts (Twilio)	Target: Q1 2027	Extends reach to users without internet access at the moment of risk
Phase 4	GIS-based hazard mapping (Mapbox) with regional visualization	Target: Q2 2027	Enables visual, region-wide risk awareness for disaster management teams